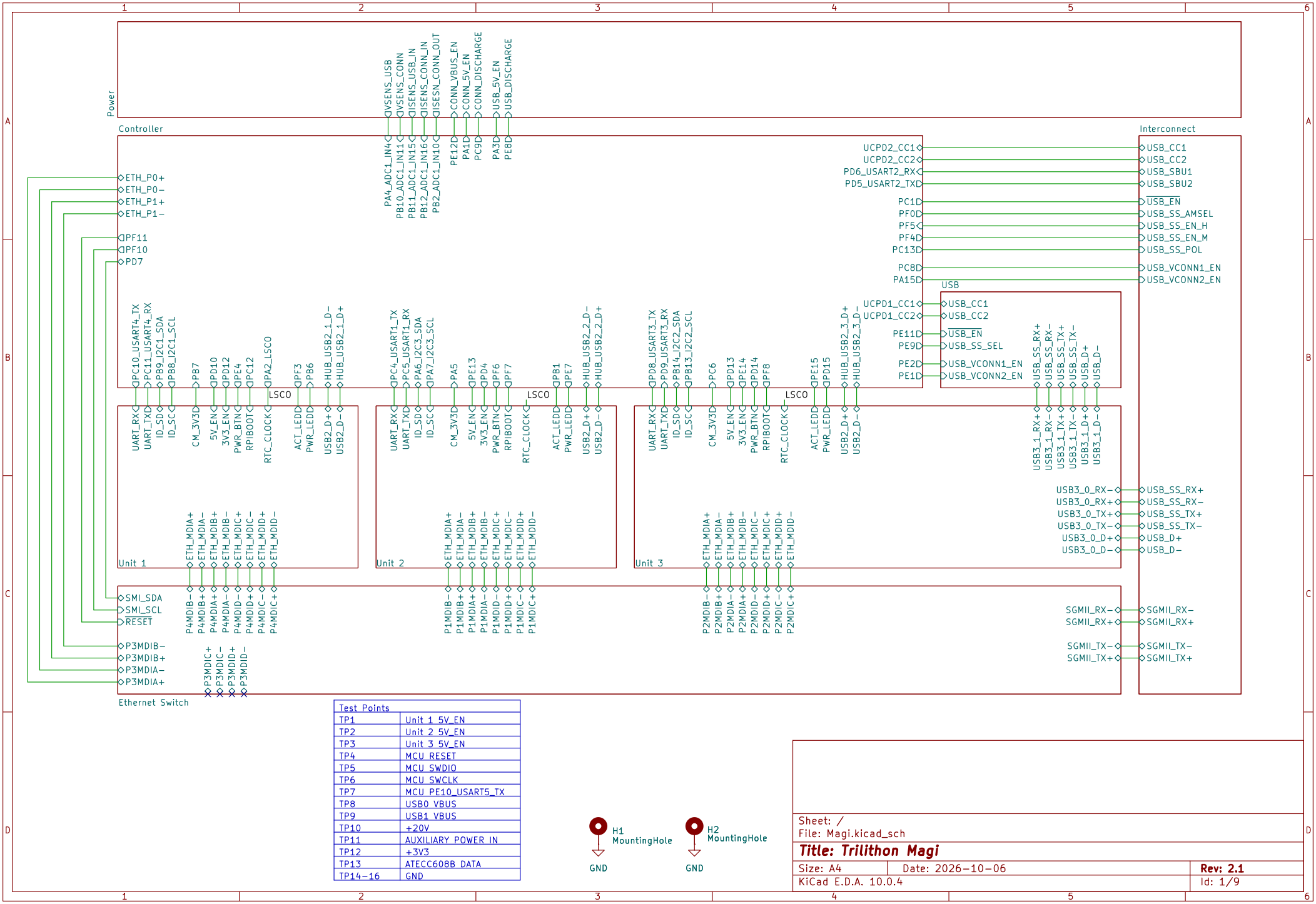


Test Points	
TP1	Unit 1 5V_EN
TP2	Unit 2 5V_EN
TP3	Unit 3 5V_EN
TP4	MCU_RESET
TP5	MCU_SWDIO
TP6	MCU_SWCLK
TP7	MCU_PE10_USART5_TX
TP8	USB0_VBUS
TP9	USB1_VBUS
TP10	+20V
TP11	AUXILIARY_POWER_IN
TP12	+3V3
TP13	ATECC608B_DATA
TP14−16	GND

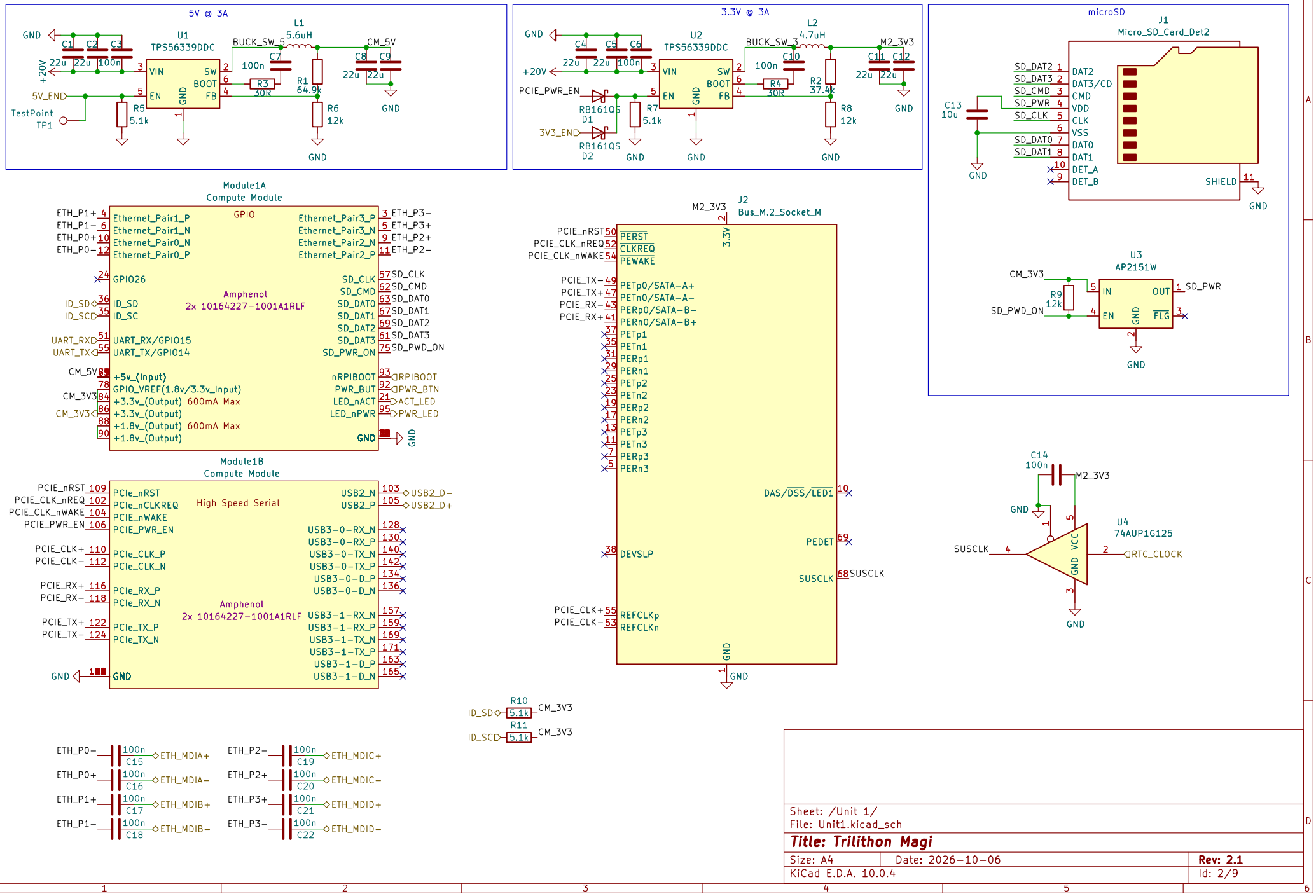
H1 MountingHole
GND

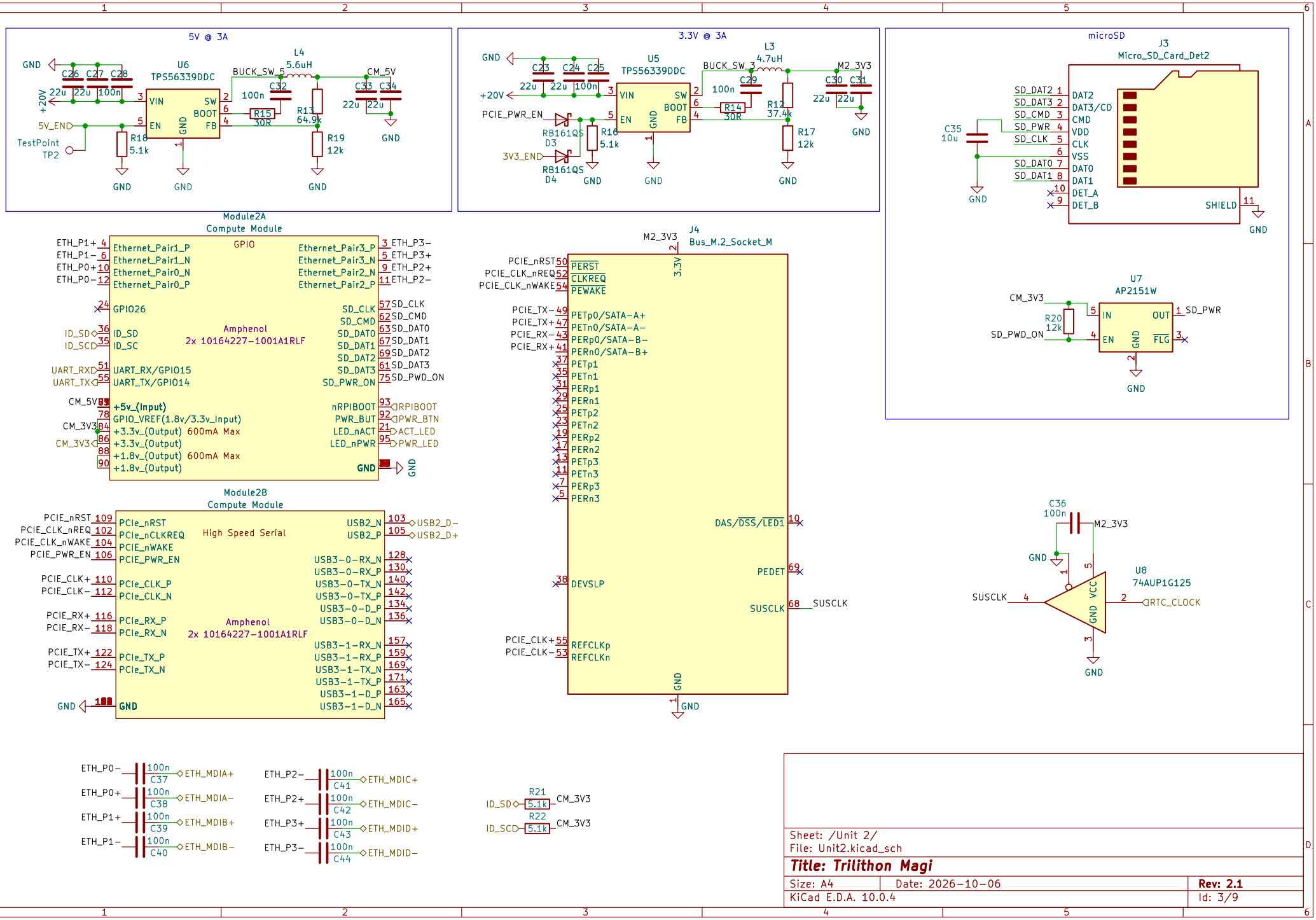
H2 MountingHole
GND

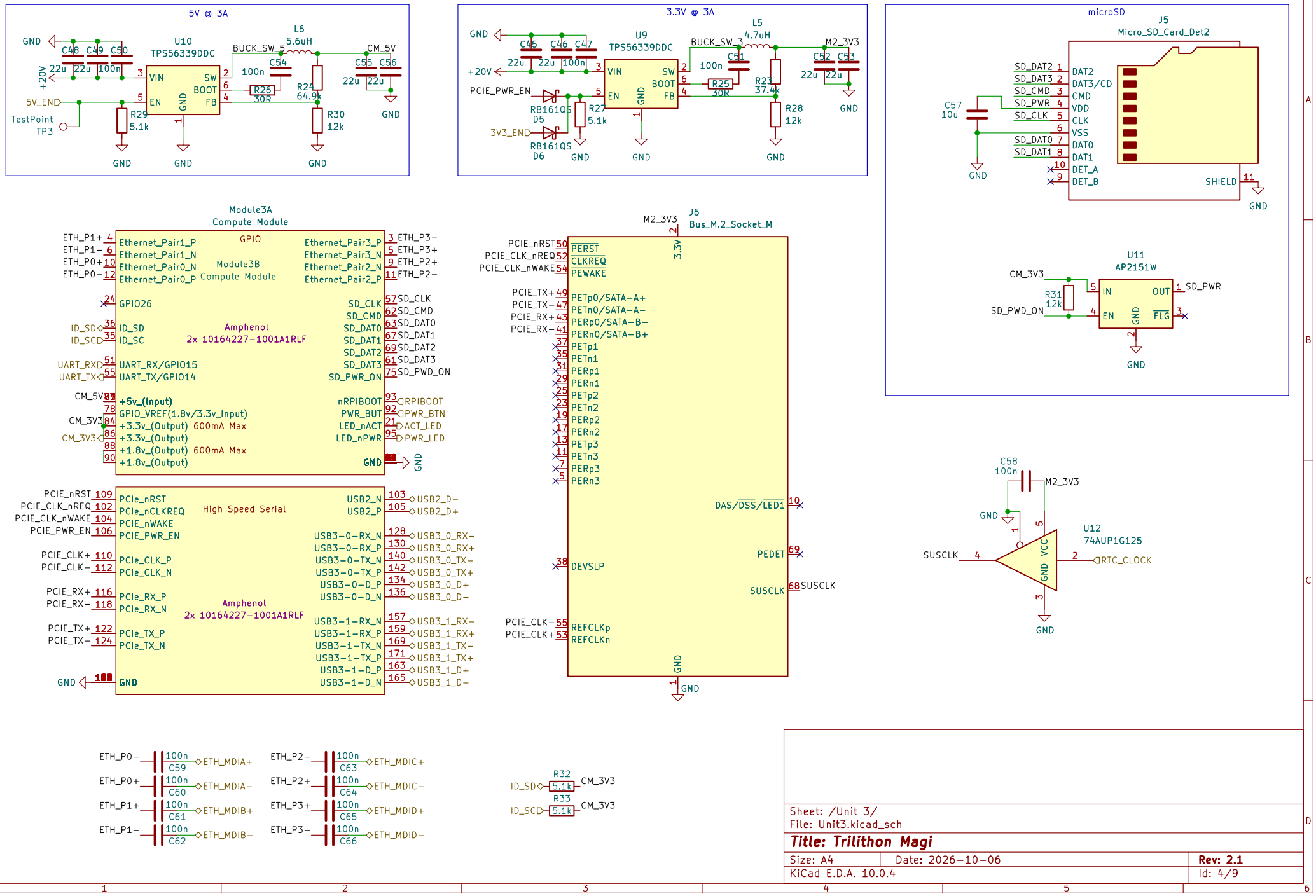
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File: Magi.kicad_sch

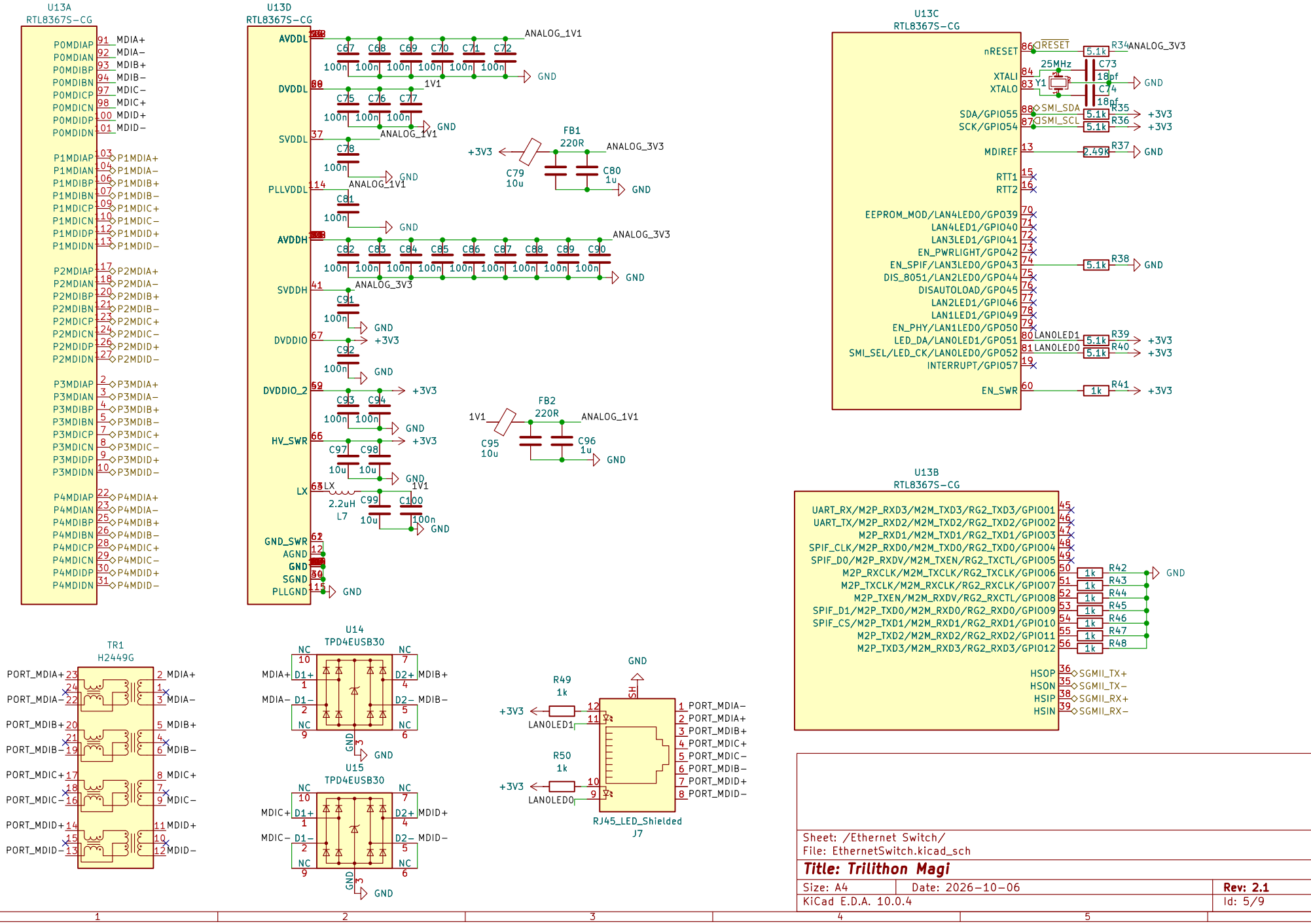
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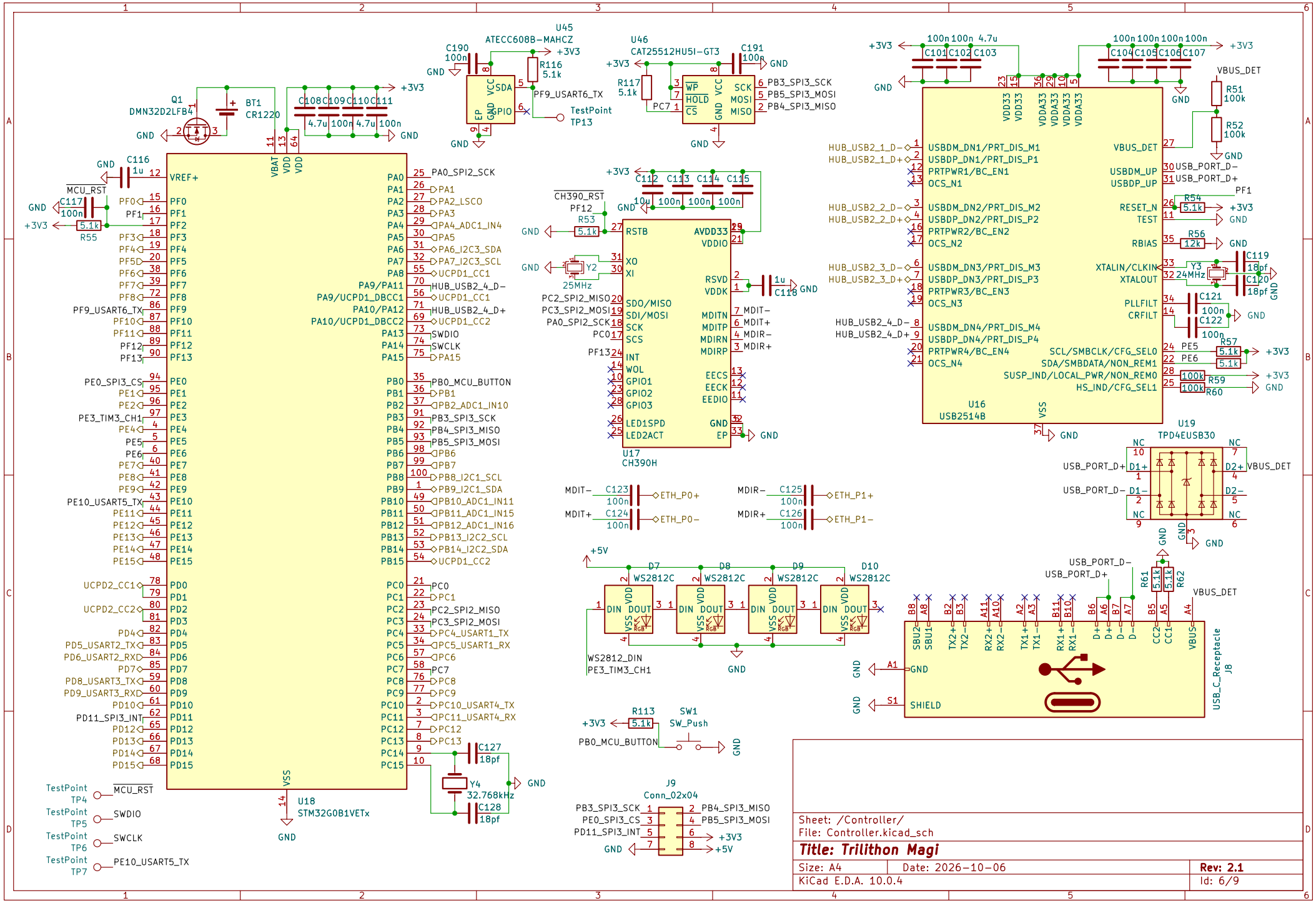
Size: A4	Date: 2026−10−06	**Rev: 2.1**
KiCad E.D.A. 10.0.4		Id: 1/9

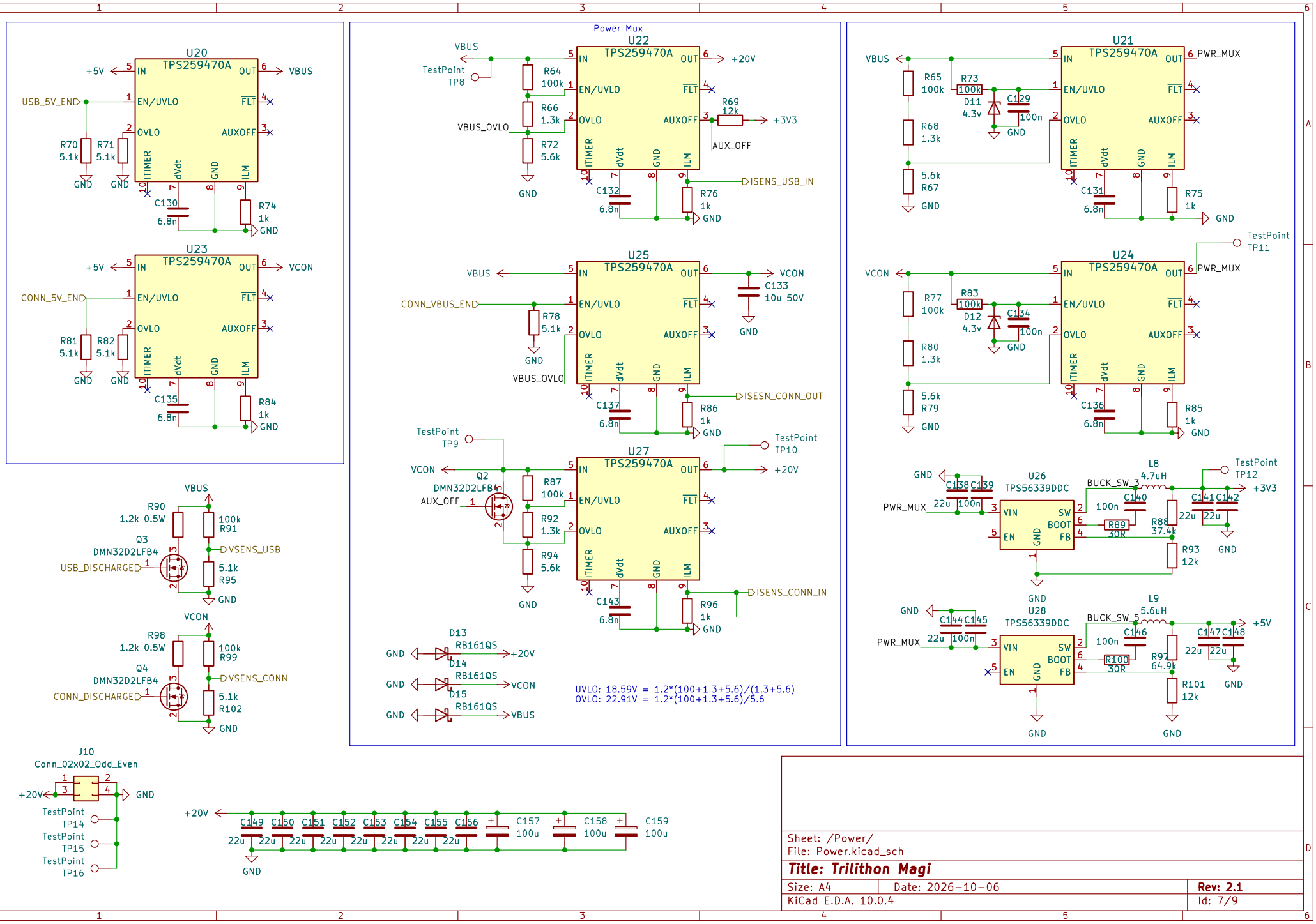

5V @ 3A
U1
TPS56339DDC
L1
5.6uH
BUCK_SW_5
CM_5V
C1 C2 C3 22u 22u 100n
C7 100n
C8 C9 22u 22u
R3 30R
R1 64.9k
R5 5.1k
R6 12k
+20V
5V_END
TestPoint
TP1
GND
3.3V @ 3A
U2
TPS56339DDC
L2
4.7uH
BUCK_SW_3
M2_3V3
C4 C5 C6 22u 22u 100n
C10 100n
C11 C12 22u 22u
R4 30R
R2 37.4k
R8 12k
R7 5.1k
PCIE_PWR_EN
RB161QS
D1
D2
3V3_END
microSD
J1
Micro_SD_Card_Det2
SD_DAT2
SD_DAT3
SD_CMD
SD_PWR
SD_CLK
SD_DAT0
SD_DAT1
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
C13 10u
U3
AP2151W
CM_3V3
R9 12k
SD_PWD_ON
IN
OUT
EN
GND
FLG
SD_PWR
Module1A
Compute Module
GPIO
Ethernet_Pair1_P
Ethernet_Pair1_N
Ethernet_Pair0_N
Ethernet_Pair0_P
Ethernet_Pair3_P
Ethernet_Pair3_N
Ethernet_Pair2_N
Ethernet_Pair2_P
ETH_P1+
ETH_P1−
ETH_P0+
ETH_P0−
ETH_P3−
ETH_P3+
ETH_P2+
ETH_P2−
GPIO26
Amphenol
2x 10164227-1001A1RLF
ID_SD
ID_SC
UART_RX/GPIO15
UART_TX/GPIO14
UART_RXD
UART_TX
SD_CLK
SD_CMD
SD_DAT0
SD_DAT1
SD_DAT2
SD_DAT3
SD_PWR_ON
+5v_(Input)
GPIO_VREF(1.8v/3.3v_Input)
+3.3v_(Output) 600mA Max
+3.3v_(Output)
+1.8v_(Output) 600mA Max
+1.8v_(Output)
nRPIBOOT
PWR_BUT
LED_nACT
LED_nPWR
RPIBOOT
PWR_BTN
ACT_LED
PWR_LED
Module1B
Compute Module
High Speed Serial
PCIe_nRST
PCIe_nCLKREQ
PCIe_nWAKE
PCIe_PWR_EN
PCIe_CLK_P
PCIe_CLK_N
PCIe_RX_P
PCIe_RX_N
PCIe_TX_P
PCIe_TX_N
USB2_N
USB2_P
USB2_D−
USB2_D+
USB3-0-RX_N
USB3-0-RX_P
USB3-0-TX_N
USB3-0-TX_P
USB3-0-D_P
USB3-0-D_N
USB3-1-RX_N
USB3-1-RX_P
USB3-1-TX_N
USB3-1-TX_P
USB3-1-D_P
USB3-1-D_N
J2
Bus_M.2_Socket_M
PERST
CLKREQ
PEWAKE
PETp0/SATA-A+
PETn0/SATA-A−
PERp0/SATA-B−
PERn0/SATA-B+
PETp1
PETn1
PERp1
PERn1
PETp2
PETn2
PERp2
PERn2
PETp3
PETn3
PERp3
PERn3
DAS/DSS/LED1
PEDET
DEVSLP
SUSCLK
REFCLKp
REFCLKn
C14 100n
U4
74AUP1G125
SUSCLK
RTC_CLOCK
R10 5.1k
R11 5.1k
ID_SD
ID_SC
C15 C16 C17 C18 100n
C19 C20 C21 C22 100n
ETH_MDIA+
ETH_MDIA−
ETH_MDIB+
ETH_MDIB−
ETH_MDIC+
ETH_MDIC−
ETH_MDID+
ETH_MDID−
Sheet: /Unit 1/
File: Unit1.kicad_sch
Title: Trilithon Magi
Size: A4
Date: 2026-10-06
Rev: 2.1
KiCad E.D.A. 10.0.4
Id: 2/9

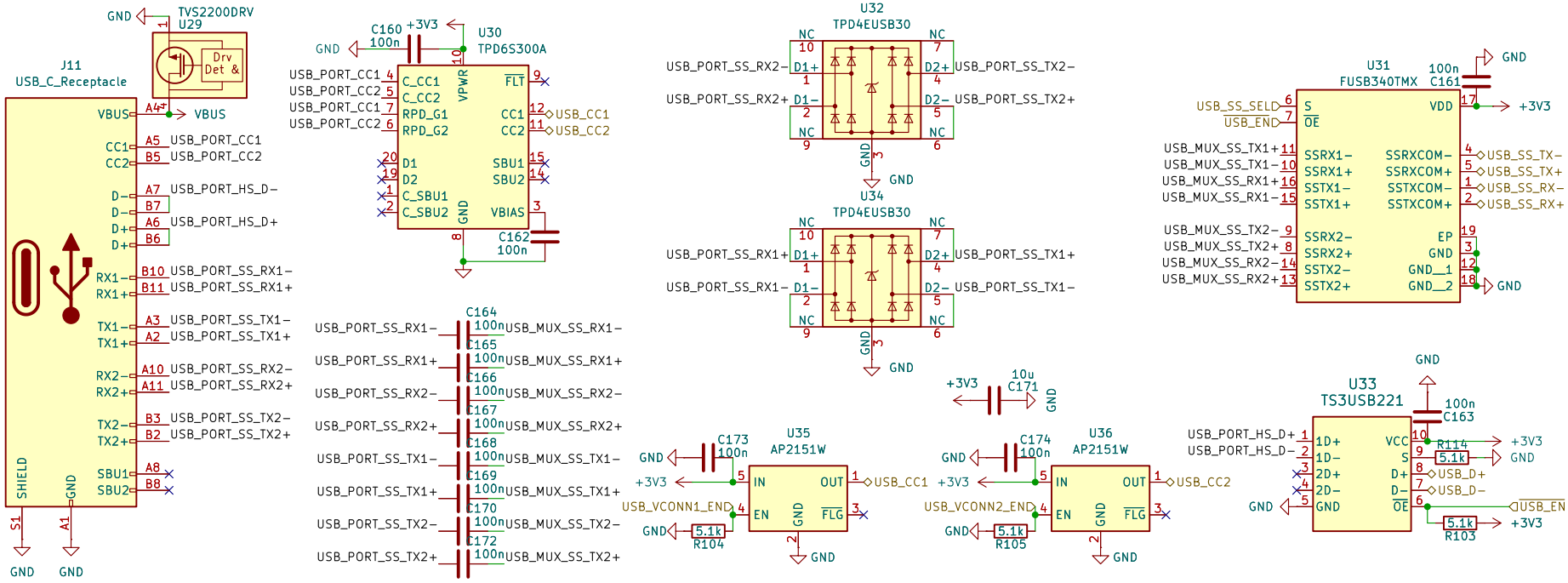

5V @ 3A
U6
TPS56339DDC
L4
5.6uH
BUCK_SW_5
CM_5V
C26 C27 C28
22u 22u 100n
C32
100n
C33 C34
22u 22u
+20V
5V_END
TestPoint
TP2
R18
5.1k
R15
30R
R13
64.9k
R19
12k
GND
3.3V @ 3A
U5
TPS56339DDC
L3
4.7uH
BUCK_SW_3
M2_3V3
C23 C24 C25
22u 22u 100n
C29
100n
C30 C31
22u 22u
+20V
PCIE_PWR_EN
3V3_END
RB161QS
D3
RB161QS
D4
R16
5.1k
R14
30R
R12
37.4k
R17
12k
microSD
J3
Micro_SD_Card_Det2
C35
10u
SD_DAT2
SD_DAT3
SD_CMD
SD_PWR
SD_CLK
SD_DAT0
SD_DAT1
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
Module2A
Compute Module
GPIO
Amphenol
2x 10164227-1001A1RLF
ETH_P1+
ETH_P1−
ETH_P0+
ETH_P0−
Ethernet_Pair1_P
Ethernet_Pair1_N
Ethernet_Pair0_N
Ethernet_Pair0_P
Ethernet_Pair3_P
Ethernet_Pair3_N
Ethernet_Pair2_N
Ethernet_Pair2_P
ETH_P3−
ETH_P3+
ETH_P2+
ETH_P2−
GPIO26
ID_SD
ID_SC
UART_RX/GPIO15
UART_TX/GPIO14
SD_CLK
SD_CMD
SD_DAT0
SD_DAT1
SD_DAT2
SD_DAT3
SD_PWR_ON
+5v_(Input)
GPIO_VREF(1.8v/3.3v_Input)
+3.3v_(Output) 600mA Max
+3.3v_(Output)
+1.8v_(Output) 600mA Max
+1.8v_(Output)
nRPIBOOT
PWR_BUT
LED_nACT
LED_nPWR
RPIBOOT
PWR_BTN
ACT_LED
PWR_LED
Module2B
Compute Module
High Speed Serial
Amphenol
2x 10164227-1001A1RLF
PCIE_nRST
PCIE_CLK_nREQ
PCIE_CLK_nWAKE
PCIE_PWR_EN
PCIe_nRST
PCIe_nCLKREQ
PCIE_nWAKE
PCIE_PWR_EN
PCIE_CLK+
PCIE_CLK−
PCIE_RX+
PCIE_RX−
PCIE_TX+
PCIE_TX−
PCIe_CLK_P
PCIe_CLK_N
PCIe_RX_P
PCIe_RX_N
PCIe_TX_P
PCIe_TX_N
USB2_N
USB2_P
USB2_D−
USB2_D+
USB3−0−RX_N
USB3−0−RX_P
USB3−0−TX_N
USB3−0−TX_P
USB3−0−D_P
USB3−0−D_N
USB3−1−RX_N
USB3−1−RX_P
USB3−1−TX_N
USB3−1−TX_P
USB3−1−D_P
USB3−1−D_N
J4
Bus_M.2_Socket_M
M2_3V3
PCIE_nRST
PCIE_CLK_nREQ
PCIE_CLK_nWAKE
PERST
CLKREQ
PEWAKE
PCIE_TX−
PCIE_TX+
PCIE_RX−
PCIE_RX+
PETp0/SATA−A+
PETn0/SATA−A−
PERp0/SATA−B−
PERn0/SATA−B+
PETp1
PETn1
PERp1
PERn1
PETp2
PETn2
PERp2
PERn2
PETp3
PETn3
PERp3
PERn3
DAS/DSS/LED1
PEDET
DEVSLP
SUSCLK
PCIE_CLK+
PCIE_CLK−
REFCLKp
REFCLKn
U7
AP2151W
CM_3V3
R20
12k
SD_PWD_ON
IN
OUT
EN
GND
FLG
SD_PWR
C36
100n
U8
74AUP1G125
SUSCLK
VCC
RTC_CLOCK
ETH_P0−
ETH_P0+
ETH_P1+
ETH_P1−
ETH_P2−
ETH_P2+
ETH_P3+
ETH_P3−
C37
C38
C39
C40
C41
C42
C43
C44
ETH_MDIA+
ETH_MDIA−
ETH_MDIB+
ETH_MDIB−
ETH_MDIC+
ETH_MDIC−
ETH_MDID+
ETH_MDID−
ID_SD
ID_SC
R21
R22
5.1k
CM_3V3
Sheet: /Unit 2/
File: Unit2.kicad_sch
Title: Trilithon Magi
Size: A4
Date: 2026−10−06
Rev: 2.1
KiCad E.D.A. 10.0.4
Id: 3/9


5V @ 3A
U10
TPS56339DDC
L6
5.6uH
BUCK_SW_5
CM_5V
C48 C49 C50
22u 22u 100n
C54
100n
C55 C56
22u 22u
R26
30R
R24
64.9k
R29
5.1k
R30
12k
+20V
5V_EN
TestPoint
TP3
GND
3.3V @ 3A
U9
TPS56339DDC
L5
4.7uH
BUCK_SW_3
M2_3V3
C45 C46 C47
C51
C52 C53
R25
30R
R23
37.4k
R27
5.1k
R28
12k
PCIE_PWR_EN
RB161QS
D5
D6
3V3_END
microSD
J5
Micro_SD_Card_Det2
C57
10u
SD_DAT2
SD_DAT3
SD_CMD
SD_PWR
SD_CLK
SD_DAT0
SD_DAT1
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
U11
AP2151W
CM_3V3
R31
12k
SD_PWD_ON
IN
OUT
EN
GND
FLG
SD_PWR
Module3A
Compute Module
GPIO
Module3B
Compute Module
Amphenol
2x 10164227-1001A1RLF
ETH_P1+
ETH_P1−
ETH_P0+
ETH_P0−
Ethernet_Pair1_P
Ethernet_Pair1_N
Ethernet_Pair0_N
Ethernet_Pair0_P
Ethernet_Pair3_P
Ethernet_Pair3_N
Ethernet_Pair2_N
Ethernet_Pair2_P
ETH_P3−
ETH_P3+
ETH_P2+
ETH_P2−
GPIO26
ID_SD
ID_SC
UART_RX/GPIO15
UART_TX/GPIO14
SD_CLK
SD_CMD
SD_DAT0
SD_DAT1
SD_DAT2
SD_DAT3
SD_PWR_ON
SD_PWD_ON
+5v_(Input)
GPIO_VREF(1.8v/3.3v_Input)
+3.3v_(Output) 600mA Max
+1.8v_(Output) 600mA Max
nRPIBOOT
PWR_BUT
LED_nACT
LED_nPWR
RPIBOOT
PWR_BTN
ACT_LED
PWR_LED
GND
PCIE_nRST
PCIE_CLK_nREQ
PCIE_CLK_nWAKE
PCIE_PWR_EN
PCIE_CLK+
PCIE_CLK−
PCIE_RX+
PCIE_RX−
PCIE_TX+
PCIE_TX−
High Speed Serial
PCIe_nRST
PCIe_nCLKREQ
PCIE_nWAKE
PCIE_PWR_EN
PCIe_CLK_P
PCIe_CLK_N
PCIe_RX_P
PCIe_RX_N
PCIe_TX_P
PCIe_TX_N
USB2_N
USB2_P
USB3-0-RX_N
USB3-0-RX_P
USB3-0-TX_N
USB3-0-TX_P
USB3-0-D_P
USB3-0-D_N
USB3-1-RX_N
USB3-1-RX_P
USB3-1-TX_N
USB3-1-TX_P
USB3-1-D_P
USB3-1-D_N
USB2_D−
USB2_D+
USB3_0_RX−
USB3_0_RX+
USB3_0_TX−
USB3_0_TX+
USB3_0_D+
USB3_0_D−
USB3_1_RX−
USB3_1_RX+
USB3_1_TX−
USB3_1_TX+
USB3_1_D+
USB3_1_D−
J6
Bus_M.2_Socket_M
M2_3V3
3.3V
PERST
CLKREQ
PEWAKE
PETp0/SATA-A+
PETn0/SATA-A−
PERp0/SATA-B−
PERn0/SATA-B+
PETp1
PETn1
PERp1
PERn1
PETp2
PETn2
PERp2
PERn2
PETp3
PETn3
PERp3
PERn3
DEVSLP
REFCLKp
REFCLKn
DAS/DSS/LED1
PEDET
SUSCLK
C58
100n
U12
74AUP1G125
SUSCLK
RTC_CLOCK
VCC
ETH_P0−
ETH_P0+
ETH_P1+
ETH_P1−
ETH_P2−
ETH_P2+
ETH_P3+
ETH_P3−
C59
C60
C61
C62
C63
C64
C65
C66
ETH_MDIA+
ETH_MDIA−
ETH_MDIB+
ETH_MDIB−
ETH_MDIC+
ETH_MDIC−
ETH_MDID+
ETH_MDID−
R32
R33
ID_SD
ID_SCD
5.1k
Sheet: /Unit 3/
File: Unit3.kicad_sch
Title: Trilithon Magi
Size: A4
Date: 2026-10-06
Rev: 2.1
KiCad E.D.A. 10.0.4
Id: 4/9

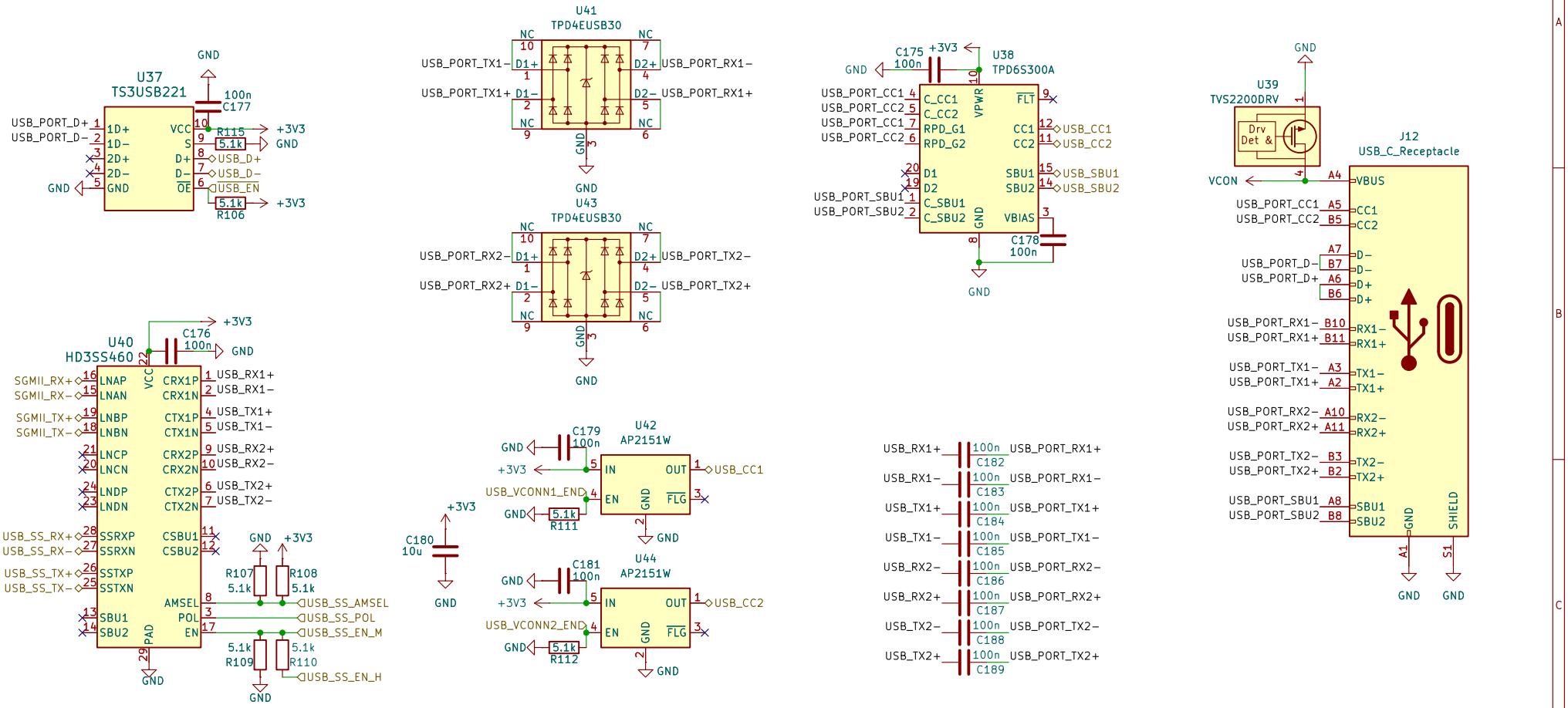

U13A
RTL8367S-CG
P0MDIAP 91 MDIA+
P0MDIAN 92 MDIA−
P0MDIBP 93 MDIB+
P0MDIBN 94 MDIB−
P0MDICP 97 MDIC−
P0MDICN 98 MDIC+
P0MDIDP 100 MDID+
P0MDIDN 101 MDID−
P1MDIAP 03 P1MDIA+
P1MDIAN 04 P1MDIA−
P1MDIBP 06 P1MDIB+
P1MDIBN 07 P1MDIB−
P1MDICP 09 P1MDIC+
P1MDICN 10 P1MDIC−
P1MDIDP 12 P1MDID+
P1MDIDN 13 P1MDID−
P2MDIAP 17 P2MDIA+
P2MDIAN 18 P2MDIA−
P2MDIBP 20 P2MDIB+
P2MDIBN 21 P2MDIB−
P2MDICP 23 P2MDIC+
P2MDICN 24 P2MDIC−
P2MDIDP 26 P2MDID+
P2MDIDN 27 P2MDID−
P3MDIAP 2 P3MDIA+
P3MDIAN 3 P3MDIA−
P3MDIBP 4 P3MDIB+
P3MDIBN 5 P3MDIB−
P3MDICP 7 P3MDIC+
P3MDICN 8 P3MDIC−
P3MDIDP 9 P3MDID+
P3MDIDN 10 P3MDID−
P4MDIAP 22 P4MDIA+
P4MDIAN 23 P4MDIA−
P4MDIBP 25 P4MDIB+
P4MDIBN 26 P4MDIB−
P4MDICP 28 P4MDIC+
P4MDICN 29 P4MDIC−
P4MDIDP 30 P4MDID+
P4MDIDN 31 P4MDID−
U13D
RTL8367S-CG
AVDDL
ANALOG_1V1
C67 C68 C69 C70 C71 C72
100n
GND
DVDDL
1V1
C75 C76 C77
SVDDL
ANALOG_1V1
C78
PLLVDDL
C81
AVDDH
ANALOG_3V3
C82 C83 C84 C85 C86 C87 C88 C89 C90
SVDDH
C91
DVDDIO
+3V3
C92
DVDDIO_2
C93 C94
HV_SWR
C97 C98
10u
LX
2.2uH
L7
C99 C100
GND_SWR
AGND
GND
SGND
PLLGND
FB1
220R
+3V3
C79
10u
C80
1u
FB2
1V1
C95
10u
C96
1u
U13C
RTL8367S-CG
nRESET
RESET
R34 5.1k
ANALOG_3V3
25MHz
XTALI
XTALO
Y1
C73 18pf
C74 18pf
SDA/GPIO55
SCK/GPIO54
SMI_SDA
SMI_SCL
R35 5.1k
R36 5.1k
MDIREF
R37 2.49k
RTT1
RTT2
EEPROM_MOD/LAN4LED0/GPO39
LAN4LED1/GPIO40
LAN3LED1/GPIO41
EN_PWRLIGHT/GPO42
EN_SPIF/LAN3LED0/GPO43
R38 5.1k
DIS_8051/LAN2LED0/GPO44
DISAUTOLOAD/GPO45
LAN2LED1/GPIO46
LAN1LED1/GPIO49
EN_PHY/LAN1LED0/GPO50
LED_DA/LAN0LED1/GPO51
LAN0LED1
R39 5.1k
SMI_SEL/LED_CK/LAN0LED0/GPO52
LAN0LED0
R40 5.1k
INTERRUPT/GPIO57
EN_SWR
R41 1k
U13B
RTL8367S-CG
UART_RX/M2P_RXD3/M2M_TXD3/RG2_TXD3/GPIO01
UART_TX/M2P_RXD2/M2M_TXD2/RG2_TXD2/GPIO02
M2P_RXD1/M2M_TXD1/RG2_TXD1/GPIO03
SPIF_CLK/M2P_RXD0/M2M_TXD0/RG2_TXD0/GPIO04
SPIF_DO/M2P_RXDV/M2M_TXEN/RG2_TXCTL/GPIO05
M2P_RXCLK/M2M_TXCLK/RG2_TXCLK/GPIO06
M2P_TXCLK/M2M_RXCLK/RG2_RXCLK/GPIO07
M2P_TXEN/M2M_RXDV/RG2_RXCTL/GPIO08
SPIF_D1/M2P_TXD0/M2M_RXD0/RG2_RXD0/GPIO09
SPIF_CS/M2P_TXD1/M2M_RXD1/RG2_RXD1/GPIO10
M2P_TXD2/M2M_RXD2/RG2_RXD2/GPIO11
M2P_TXD3/M2M_RXD3/RG2_RXD3/GPIO12
R42 1k
R43 1k
R44 1k
R45 1k
R46 1k
R47 1k
R48 1k
GND
HSOP SGMII_TX+
HSON SGMII_TX−
HSIP SGMII_RX+
HSIN SGMII_RX−
TR1
H2449G
PORT_MDIA+ MDIA+
PORT_MDIA− MDIA−
PORT_MDIB+ MDIB+
PORT_MDIB− MDIB−
PORT_MDIC+ MDIC+
PORT_MDIC− MDIC−
PORT_MDID+ MDID+
PORT_MDID− MDID−
U14
TPD4EUSB30
MDIA+ D1+
MDIA− D1−
D2+ MDIB+
D2− MDIB−
NC
GND
U15
TPD4EUSB30
MDIC+ D1+
MDIC− D1−
D2+ MDID+
D2− MDID−
R49
1k
LAN0LED1
R50
1k
LAN0LED0
SH
PORT_MDIA−
PORT_MDIA+
PORT_MDIB+
PORT_MDIC+
PORT_MDIC−
PORT_MDIB−
PORT_MDID+
PORT_MDID−
RJ45_LED_Shielded
J7
Sheet: /Ethernet Switch/
File: EthernetSwitch.kicad_sch
Title: Trilithon Magi
Size: A4
Date: 2026-10-06
Rev: 2.1
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U45
ATECC608B-MAHCZ
C190
100n
R116
5.1k
+3V3
GND
PF9_USART6_TX
TestPoint
TP13
U46
CAT25512HU5I-GT3
C191
100n
R117
5.1k
PC7
PB3_SPI3_SCK
PB5_SPI3_MOSI
PB4_SPI3_MISO
Q1
DMN32D2LFB4
BT1
CR1220
C108
C109
C110
C111
4.7u
100n
4.7u
100n
C101
C102
C103
C104
C105
C106
C107
VBUS_DET
R51
100k
R52
100k
USB_PORT_D-
USB_PORT_D+
R54
5.1k
R56
12k
C119
18pf
C120
18pf
Y3
24MHz
C121
100n
C122
100n
R57
5.1k
R59
100k
R60
100k
PE5
PE6
PF1
MCU_RST
C116
1u
C117
100n
R55
5.1k
C112
C113
C114
C115
10u
100n
CH390_RST
R53
5.1k
Y2
25MHz
C118
1u
U16
USB2514B
U17
CH390H
U18
STM32G0B1VETx
U19
TPD4EUSB30
USB_PORT_D+
USB_PORT_D-
VBUS_DET
MDIT-
MDIT+
MDIR-
MDIR+
C123
C124
C125
C126
ETH_P0+
ETH_P0-
ETH_P1+
ETH_P1-
D7
D8
D9
D10
WS2812C
WS2812_DIN
PE3_TIM3_CH1
R61
5.1k
R62
5.1k
USB_C_Receptacle
J8
R113
5.1k
SW1
SW_Push
PB0_MCU_BUTTON
J9
Conn_02x04
C127
18pf
Y4
32.768kHz
C128
18pf
TestPoint
TP4
TP5
TP6
TP7
SWDIO
SWCLK
PE10_USART5_TX
Sheet: /Controller/
File: Controller.kicad_sch
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Power Mux
U20
TPS259470A
U23
TPS259470A
U22
TPS259470A
U25
TPS259470A
U27
TPS259470A
U21
TPS259470A
U24
TPS259470A
U26
TPS56339DDC
U28
TPS56339DDC
L8
4.7uH
L9
5.6uH
J10
Conn_02x02_Odd_Even
VBUS
VCON
+5V
+20V
+3V3
GND
USB_5V_END
CONN_5V_END
VBUS_OVLO
CONN_VBUS_END
VCON_OVLO
AUX_OFF
USB_DISCHARGED
CONN_DISCHARGED
VSENS_USB
VSENS_CONN
ISENS_USB_IN
ISESN_CONN_OUT
ISENS_CONN_IN
PWR_MUX
BUCK_SW_3
BUCK_SW_5
TestPoint TP8
TestPoint TP9
TestPoint TP10
TestPoint TP11
TestPoint TP12
TestPoint TP14
TestPoint TP15
TestPoint TP16
Q2 DMN32D2LFB4
Q3 DMN32D2LFB4
Q4 DMN32D2LFB4
D11 4.3v
D12 4.3v
D13 RB161QS
D14 RB161QS
D15 RB161QS
UVLO: 18.59V = 1.2*(100+1.3+5.6)/(1.3+5.6)
OVLO: 22.91V = 1.2*(100+1.3+5.6)/5.6
Sheet: /Power/
File: Power.kicad_sch
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Sheet: /USB/		
File: USB.kicad_sch		
Title: Trilithon Magi		
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Sheet: /Interconnect/		
File: Connector.kicad_sch		
Title: Trilithon Magi		
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